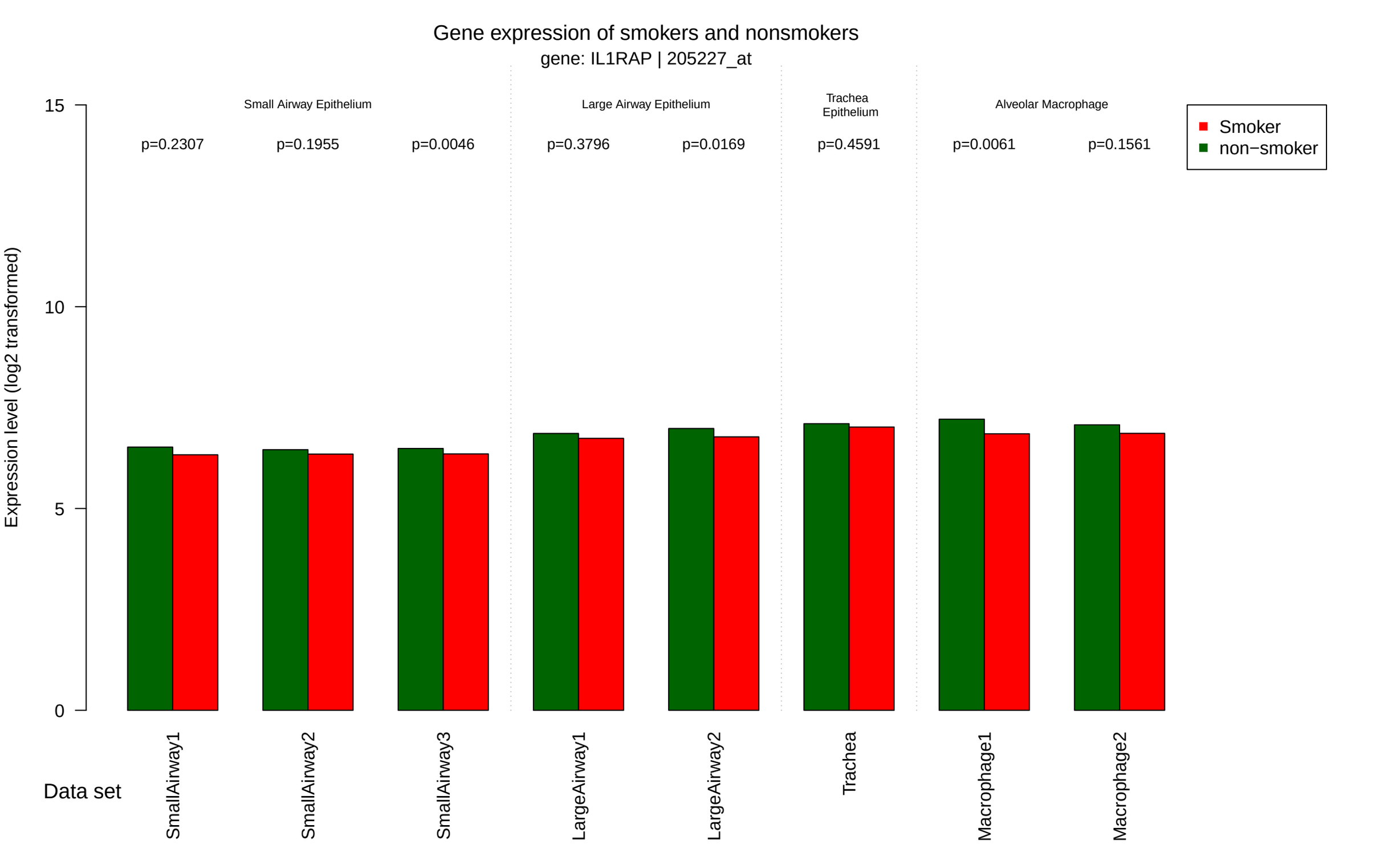
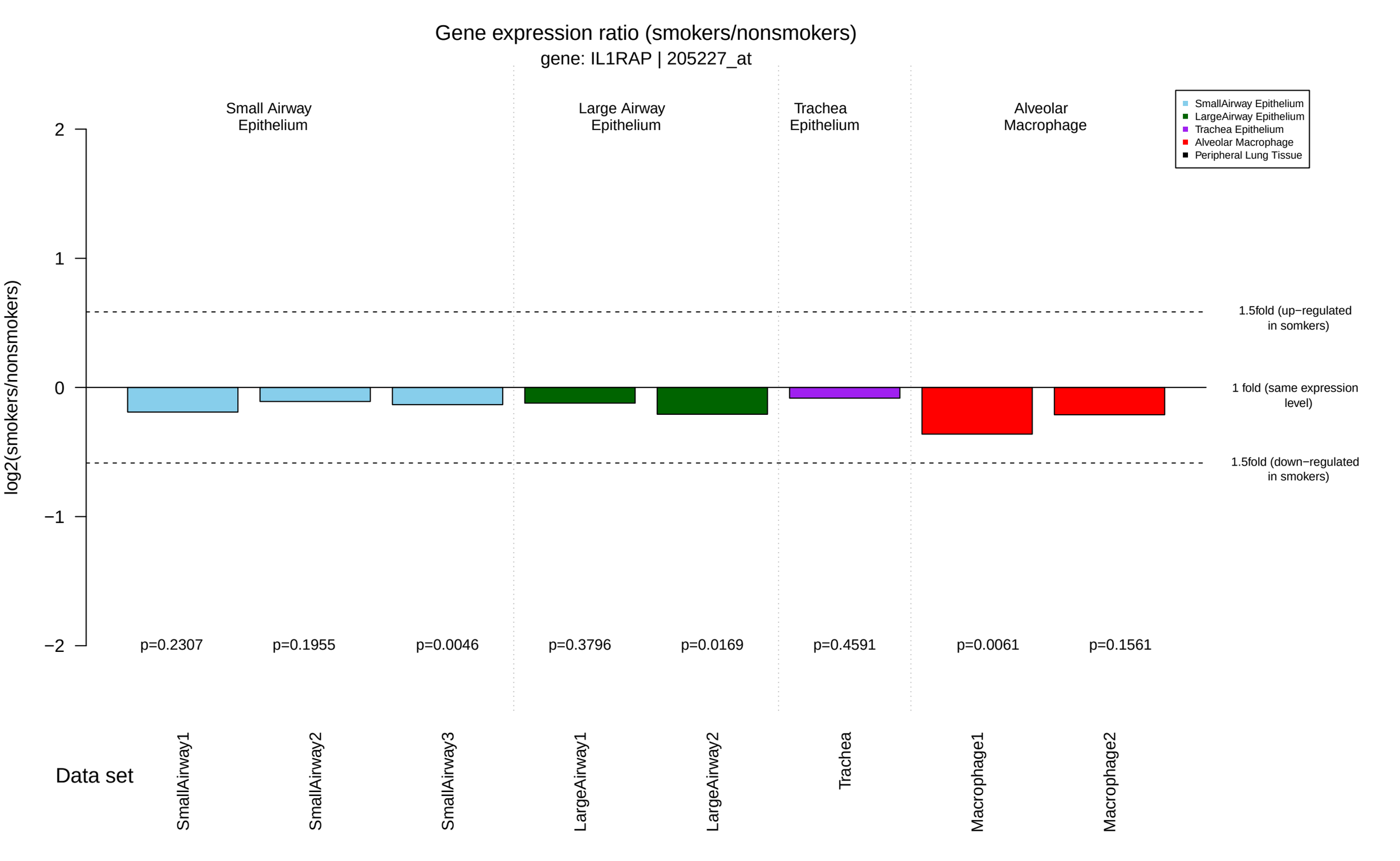
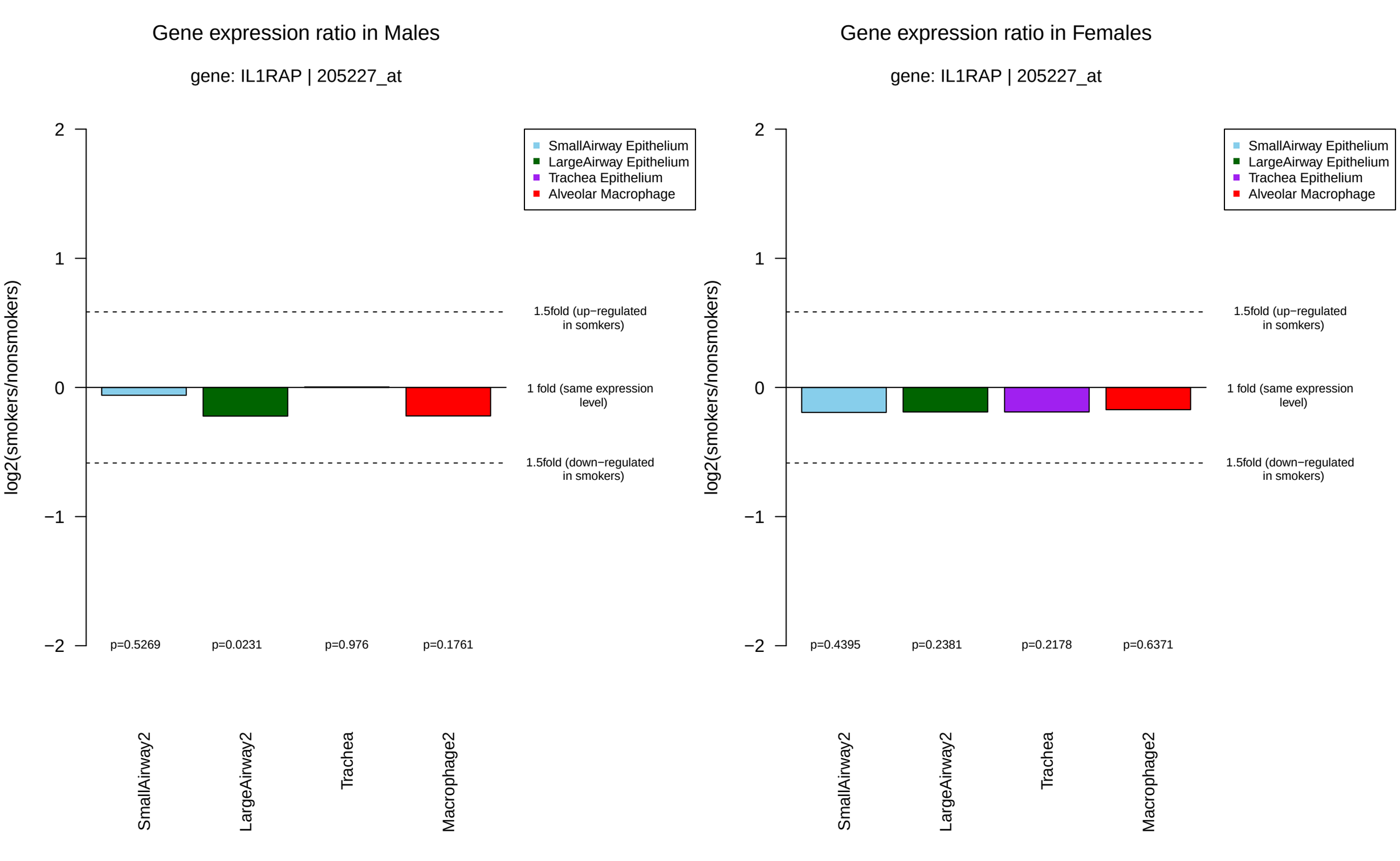
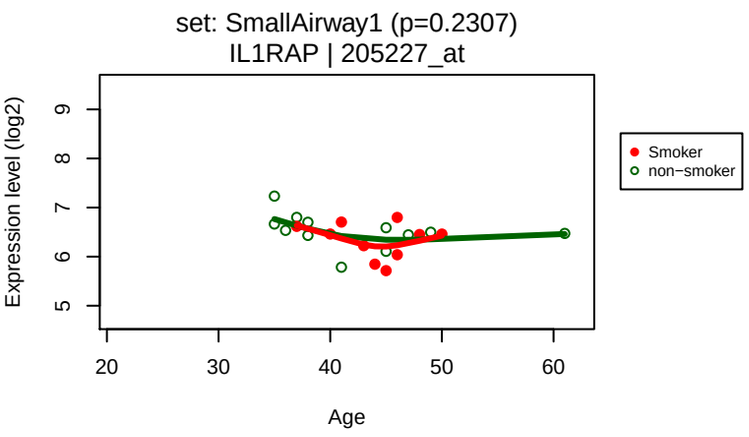
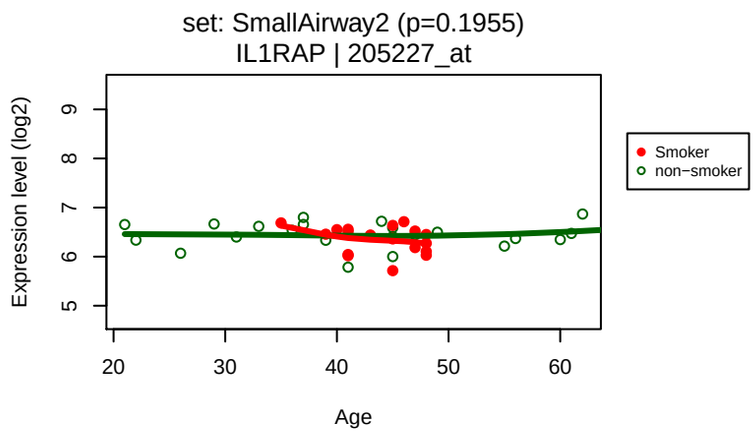
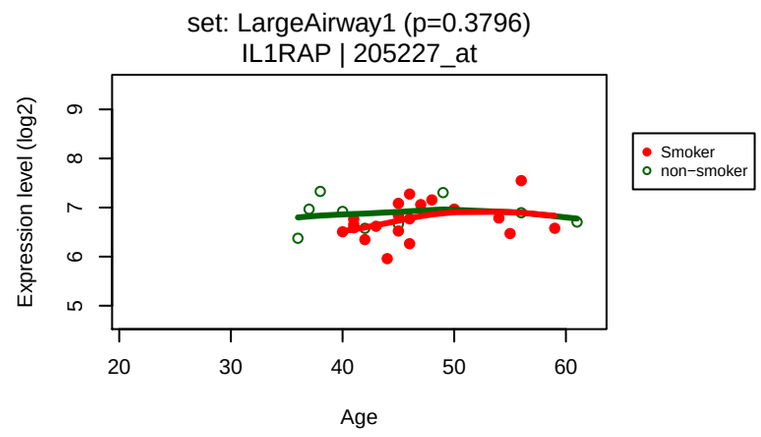

Gene expression of smokers and nonsmokers
gene: IL1RAP | 205227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2307
p=0.1955
p=0.0046
p=0.3796
p=0.0169
p=0.4591
p=0.0061
p=0.1561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

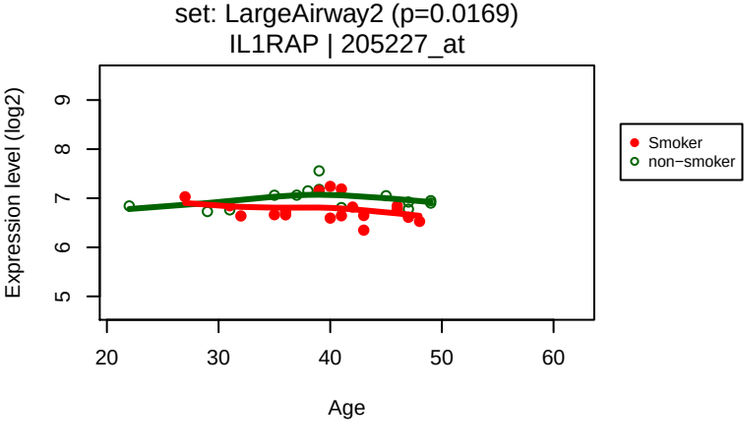

Gene expression ratio (smokers/nonsmokers)
gene: IL1RAP | 205227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2307
p=0.1955
p=0.0046
p=0.3796
p=0.0169
p=0.4591
p=0.0061
p=0.1561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

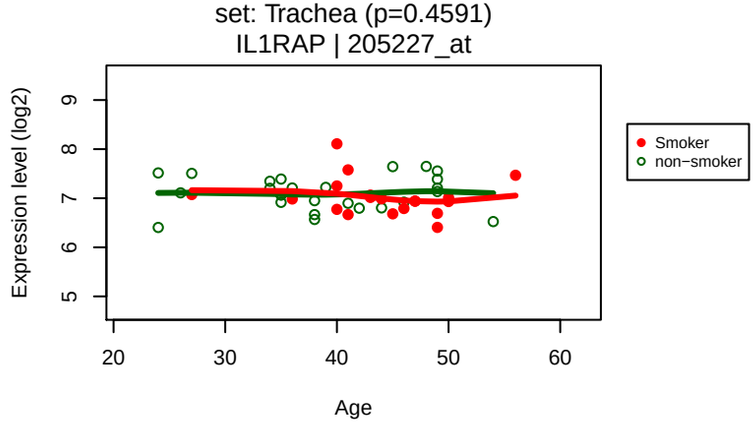

Gene expression ratio in Males
gene: IL1RAP | 205227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5269
p=0.0231
p=0.976
p=0.1761
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL1RAP | 205227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4395
p=0.2381
p=0.2178
p=0.6371
SmallAirway2
LargeAirway2
Trachea
Macrophage2

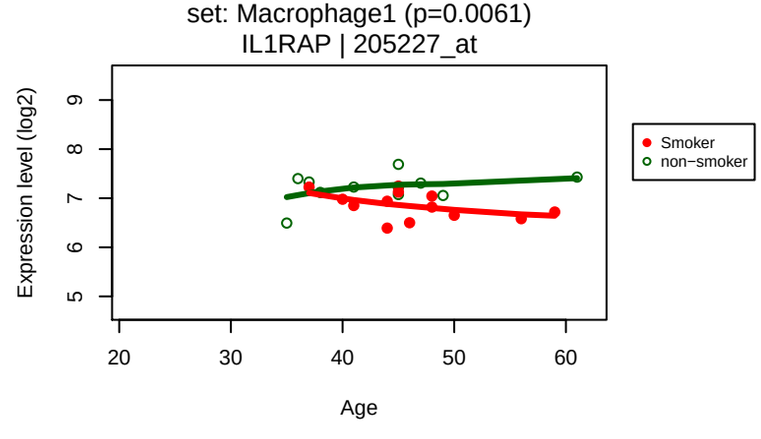

set: SmallAirway1 (p=0.2307)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker

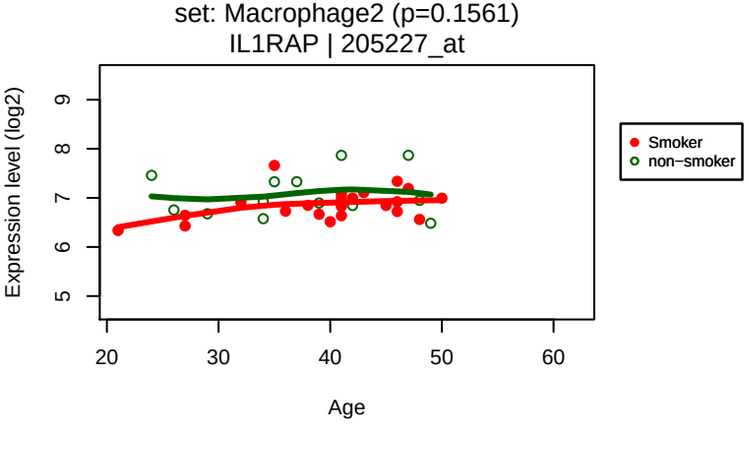

set: SmallAirway2 (p=0.1955)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker

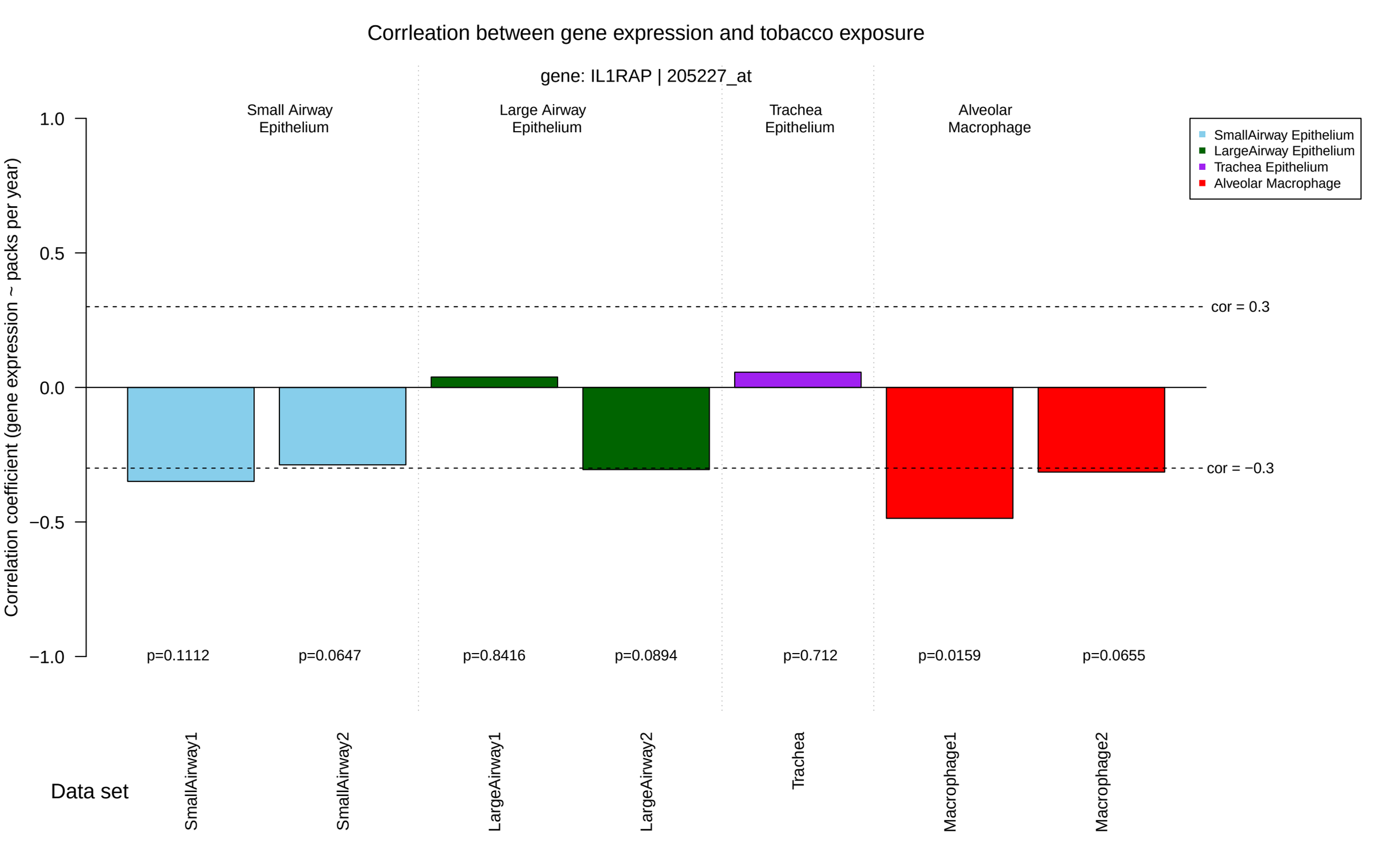

set: LargeAirway1 (p=0.3796)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0169)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4591)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0061)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1561)
IL1RAP | 205227_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL1RAP | 205227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1112
p=0.0647
p=0.8416
p=0.0894
p=0.712
p=0.0159
p=0.0655
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2